

0620 • PAPER 4

TOPIC

04

Electrochemistry

Questions only • Original examination pages retained

19

QUESTIONS

281

TOTAL MARKS

1

SECTIONS



USE THE TRACKER • ATTEMPT WITHOUT NOTES • MARK • REVIEW ERRORS

TOPIC CONTENTS

Quick navigation and progress overview

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01

Electrolysis and electrode reactions

19 questions • 281 marks

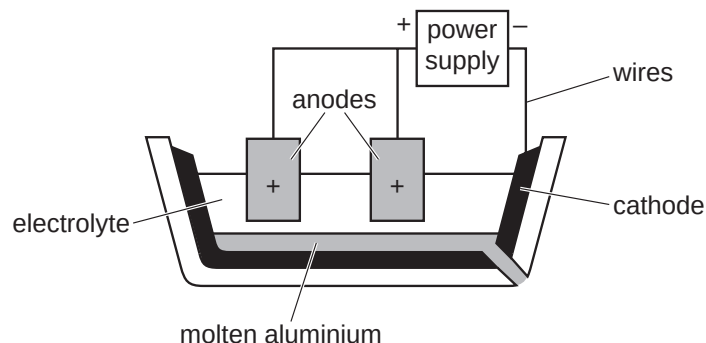
4.1 Electrolysis and electrode reactions

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01	2020 Feb/March Paper 42 Question 2	19	☐	☐	☐
02	2020 May/June Paper 42 Question 5	8	☐	☐	☐
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4.1 Electrolysis and electrode reactions

#	SOURCE	MARKS	DONE	MARK	REVIEW
15	2024 Oct/Nov Paper 41 Question 3	18	☐	☐	☐
16	2024 Oct/Nov Paper 42 Question 2	13	☐	☐	☐
17	2024 Oct/Nov Paper 43 Question 2	11	☐	☐	☐
18	2025 May/June Paper 42 Question 3	15	☐	☐	☐
19	2025 Oct/Nov Paper 43 Question 3	17	☐	☐	☐

- 2 Aluminium is extracted from its ore. The ore is converted into pure aluminium oxide, which then undergoes electrolysis as shown.



- (a) (i) Name an ore of aluminium.

..... [1]

- (ii) What is meant by the term *electrolysis*?

.....
..... [2]

- (b) Aluminium oxide has a melting point of about 2000 °C, but the electrolysis process operates at about 900 °C.

- (i) Name the compound added to aluminium oxide to reduce the operating temperature.

..... [1]

- (ii) Suggest **one** benefit to the environment of reducing the operating temperature.

.....
..... [1]

- (iii) Write the ionic half-equation for the reaction taking place at:

the negative electrode (cathode)

the positive electrode (anode)

[4]

- (iv) Explain why the anodes need frequent replacement.

.....
..... [2]

(c) Aluminium oxide reacts with acids and with alkalis.

(i) What term is used to describe an oxide that reacts with acids and with alkalis?

..... [1]

(ii) Aluminium oxide reacts with dilute sulfuric acid to form a salt.

State the name and write the formula of the salt formed.

name

formula [2]

(iii) Aluminium oxide reacts with dilute sodium hydroxide to form a salt and one other product.

Name the other product.

..... [1]

(iv) Aluminium hydroxide, $\text{Al}(\text{OH})_3$, decomposes when heated to form aluminium oxide and water.

Write the chemical equation for this reaction.

..... [2]

(v) Suggest the names of **two** other aluminium compounds that decompose when heated to form aluminium oxide.

.....

..... [2]

[Total: 19]

- 5 Electrolysis of concentrated aqueous sodium chloride using inert electrodes forms chlorine, hydrogen and sodium hydroxide.

(a) What is meant by the term *electrolysis*?

.....
.....
..... [2]

(b) Name a substance that can be used as the inert electrodes.

..... [1]

(c) Write an ionic half-equation for the formation of hydrogen during this electrolysis.

..... [1]

(d) Give the formulae of the **four** ions present in concentrated aqueous sodium chloride.

..... [2]

(e) Explain how sodium hydroxide is formed during this electrolysis.

.....
.....
..... [2]

[Total: 8]

7 Aluminium is extracted by electrolysis. Iron is extracted from its ore by reduction with carbon.

(a) What is meant by the term *electrolysis*?

.....
..... [2]

(b) Name the main ore of aluminium.

..... [1]

(c) (i) Explain why aluminium **cannot** be extracted by reduction with carbon.

..... [1]

(ii) Describe the role of cryolite in the extraction of aluminium by electrolysis.

..... [1]

(iii) Name the product formed at the positive electrode.

..... [1]

(iv) Write the ionic half-equation for the reaction at the negative electrode.

..... [2]

(d) Aluminium is used in overhead electricity cables.

Give **two** properties of aluminium that make it suitable for use in overhead electricity cables.

1

2 [2]

(e) Iron is a transition element.

(i) Iron forms hydrated iron(III) oxide when it rusts.

Write a word equation to represent the formation of rust.

..... [2]

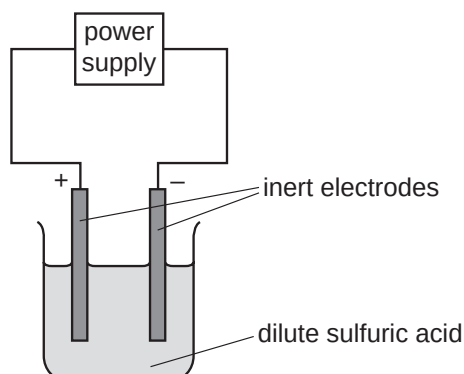
(ii) Give **two** ways in which the properties of transition elements differ from the properties of Group I metals.

1

2 [2]

[Total: 14]

- 5 (a) Dilute sulfuric acid is electrolysed using the apparatus shown in the diagram.



- (i) State what is meant by the term *electrolysis*.

.....
.....
..... [2]

- (ii) Explain why inert electrodes are used.

.....
..... [1]

- (iii) Name the products formed at each electrode.

negative electrode
positive electrode [2]

- (iv) Write an ionic half-equation for the reaction at the negative electrode.

..... [2]

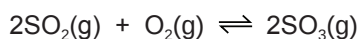
(b) Sulfuric acid is manufactured using the Contact process. This manufacture involves four stages.

(i) **Stage 1** involves the combustion of sulfur to form sulfur dioxide.

Write the chemical equation for **stage 1**.

..... [1]

(ii) The equation for **stage 2** is shown.

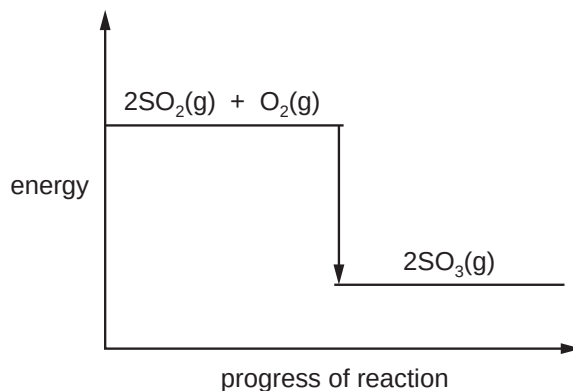


The reaction can reach equilibrium.

Explain what is meant by the term *equilibrium*.

.....
.....
.....
..... [2]

(iii) The energy level diagram for the forward reaction in **stage 2** is shown.



Explain what the diagram shows about the energy changes in the forward reaction.

.....
.....
.....
..... [2]

(c) In **stage 3** sulfur trioxide, SO_3 , is converted to oleum, $\text{H}_2\text{S}_2\text{O}_7$.

In **stage 4** oleum reacts to form sulfuric acid, H_2SO_4 .

State what oleum reacts with in **stage 4**.

..... [1]

(d) A sample of sulfuric acid, H_2SO_4 , has a concentration of 0.75 mol/dm^3 .

Calculate the concentration of sulfuric acid in g/dm^3 .

..... g/dm^3 [2]

[Total: 15]

3 Potassium reacts with chlorine to form potassium chloride, KCl .

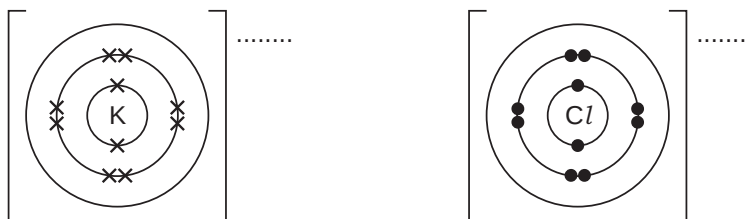
(a) Write a chemical equation for this reaction.

..... [2]

(b) Potassium chloride is an ionic compound.

Complete the diagram to show the electron arrangement in the outer shells of the ions present in potassium chloride.

Give the charges on both ions.



[3]

(c) Molten potassium chloride undergoes electrolysis.

(i) State what is meant by the term *electrolysis*.

..... [2]

(ii) Name the products formed at the positive electrode (anode) and negative electrode (cathode) when molten potassium chloride undergoes electrolysis.

anode

cathode

[2]

(d) Concentrated aqueous potassium chloride undergoes electrolysis.

(i) Write an ionic half-equation for the reaction at the negative electrode (cathode).

..... [2]

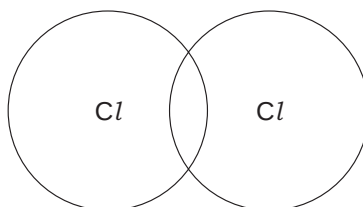
(ii) Name the product formed at the positive electrode (anode).

..... [1]

(iii) Name the potassium compound that remains in the solution after electrolysis.

..... [1]

- (e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of chlorine, Cl_2 .
Show the outer electrons only.



[1]

- (f) The melting points and boiling points of chlorine and potassium chloride are shown.

	melting point /°C	boiling point /°C
chlorine	−101	−35
potassium chloride	770	1500

- (i) Deduce the physical state of chlorine at -75°C . Use the data in the table to explain your answer.

physical state

explanation

[2]

- (ii) Explain, in terms of structure and bonding, why potassium chloride has a much higher melting point than chlorine.

Your answer should refer to the:

- types of particle held together by the forces of attraction
- types of forces of attraction between particles
- relative strength of the forces of attraction.

.....

.....

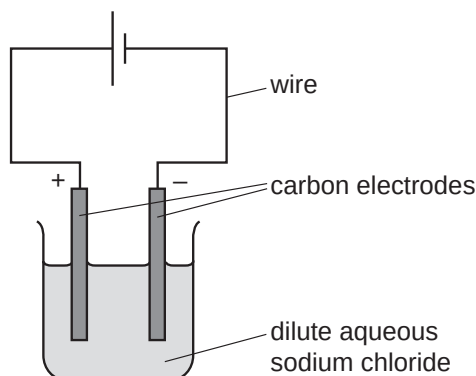
.....

.....

[3]

[Total: 19]

- 4 A student carries out an electrolysis experiment using the apparatus shown.



The student uses dilute aqueous sodium chloride.

- (a) State the name given to any solution which undergoes electrolysis.

..... [1]

- (b) Hydroxide ions are discharged at the anode.

- (i) Complete the ionic half-equation for this reaction.



- (ii) Explain how the ionic half-equation shows the hydroxide ions are being oxidised.

..... [1]

- (c) Describe what the student observes at the cathode.

..... [1]

- (d) Write the ionic half-equation for the reaction at the cathode.

..... [2]

(e) The student repeats the experiment using concentrated aqueous sodium chloride.

(i) Describe what the student observes at:

- the cathode
- the anode. [2]

(ii) The student added litmus to the solution after the electrolysis of concentrated aqueous sodium chloride.

State the colour seen in the solution. Give a reason for your answer.

colour of solution

reason [2]

(f) Carbon electrodes are used because they are inert.

State another element that can be used instead of carbon.

..... [1]

[Total: 12]

3 Sodium reacts with fluorine to form sodium fluoride, NaF.

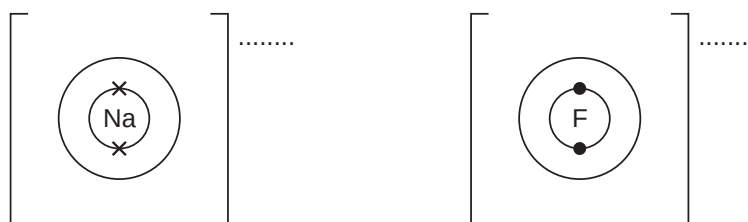
(a) Write a chemical equation for this reaction.

..... [2]

(b) Sodium fluoride is an ionic compound.

Complete the diagram to show the electron arrangement in the outer shells of the ions present in sodium fluoride.

Give the charges on both ions.



[3]

(c) Aqueous sodium fluoride undergoes electrolysis.

(i) State what is meant by the term *electrolysis*.

.....
..... [2]

(ii) Name the products formed at the positive electrode (anode) and the negative electrode (cathode) when dilute aqueous sodium fluoride undergoes electrolysis.

anode
cathode [2]

(d) Molten sodium fluoride undergoes electrolysis.

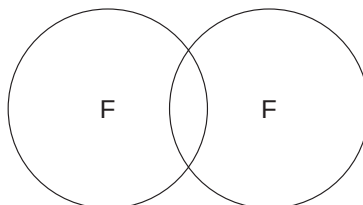
(i) Name the products formed at the positive electrode (anode) and the negative electrode (cathode) when molten sodium fluoride undergoes electrolysis.

anode
cathode [2]

(ii) Write the ionic half-equation for the reaction at the negative electrode (cathode).

..... [1]

- (e) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of fluorine, F_2 .
Show the outer electrons only.



[1]

- (f) The melting points and boiling points of fluorine and sodium fluoride are shown.

	melting point / $^{\circ}\text{C}$	boiling point / $^{\circ}\text{C}$
fluorine	-220	-188
sodium fluoride	993	1695

- (i) Deduce the physical state of fluorine at -195°C . Use the data in the table to explain your answer.

physical state

explanation

[2]

- (ii) Explain, in terms of structure and bonding, why sodium fluoride has a much higher melting point than fluorine.

Your answer should refer to the:

- types of particle held together by the forces of attraction
- types of forces of attraction between particles
- relative strength of the forces of attraction.

.....

.....

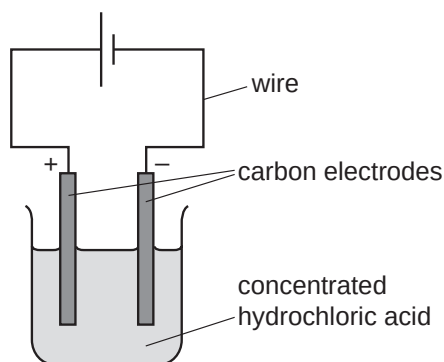
.....

..... [3]

[Total: 18]

3 This question is about electrolysis.

Concentrated hydrochloric acid is electrolysed using the apparatus shown.



(a) Chloride ions are discharged at the anode.

(i) Complete the ionic half-equation for this reaction.



(ii) State whether oxidation or reduction takes place. Explain your answer.

.....
 [1]

(b) Describe what is seen at the cathode.

..... [1]

(c) Write the ionic half-equation for the reaction at the cathode.

..... [2]

(d) The pH of the electrolyte is measured throughout the experiment.

(i) Suggest the pH of the electrolyte at the beginning of the experiment.

..... [1]

(ii) State how the pH changes, if at all, during the experiment.

Explain your answer.

.....
 [2]

(e) The electrolysis is repeated using molten lead(II) bromide.

Describe what is seen at the:

- cathode
 - anode.
- [2]

(f) State **two** properties of graphite (carbon) which make it suitable for use as an electrode.

- 1
 - 2
- [2]

[Total: 13]

2 This question is about electrolysis.

(a) State the meaning of the term *electrolyte*.

.....
 [2]

(b) The table gives information about the electrolysis of two electrolytes. Carbon (graphite) electrodes are used in each experiment.

(i) Complete the table to show the observations and products of electrolysis.

electrolyte	positive electrode (anode)		negative electrode (cathode)	
	observations	name of product	observations	name of product
aqueous copper(II) sulfate	colourless bubbles			
concentrated aqueous sodium bromide			colourless bubbles	hydrogen

[5]

(ii) Hydrogen is produced at the negative electrode (cathode) during the electrolysis of concentrated aqueous sodium bromide.

Write the ionic half-equation for this reaction.

..... [2]

(iii) State **two** reasons why carbon (graphite) is suitable to use as an electrode.

1

2 [2]

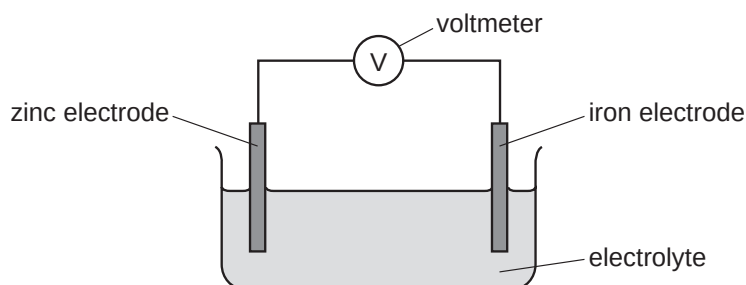
(iv) Name the particle responsible for the conduction of electricity in the metal wires used in a circuit.

..... [1]

[Total: 12]

4 This question is about chemical reactions and electricity.

(a) The diagram shows the apparatus used in the production of electrical energy in a simple cell.



The zinc electrode dissolves in the electrolyte forming $\text{Zn}^{2+}(\text{aq})$ ions.

(i) Draw an arrow on the diagram to show the direction of electron flow. [1]

(ii) Write the ionic half-equation for the reaction that occurs when the zinc electrode dissolves. [2]

.....

(b) The reading on the voltmeter can be increased if either zinc or iron is replaced by another metal.

(i) Name a metal that can replace zinc and increase the reading on the voltmeter.

..... [1]

(ii) Name a metal that can replace iron and increase the reading on the voltmeter.

..... [1]

(c) Fuel cells are used to generate electricity.

(i) Name the reactants in a fuel cell.

..... [1]

(ii) Name the waste product of a fuel cell.

..... [1]

(d) Electricity can be used to break down aqueous or molten ionic compounds.

(i) Name the process which uses electricity to break down aqueous or molten ionic compounds.

..... [1]

(ii) Explain why the ionic compound needs to be aqueous or molten.

..... [1]

(e) Brine is concentrated aqueous sodium chloride.

(i) Name **three** substances which are manufactured by passing electricity through brine.

1

2

3

[3]

(ii) Name a different substance formed when molten sodium chloride is used instead of concentrated aqueous sodium chloride.

..... [1]

[Total: 13]

5 This question is about copper and its compounds.

(a) Describe the bonding in a metallic element such as copper.

You may include a diagram as part of your answer.

.....
.....
..... [3]

(b) A metal spoon is electroplated with copper.

State what is used as:

the positive electrode (anode)

the negative electrode (cathode)

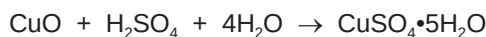
the electrolyte.

[3]

(c) The formula for crystals of hydrated copper(II) sulfate is $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

Hydrated copper(II) sulfate is made by reacting copper(II) oxide with dilute sulfuric acid.

The overall equation is shown.



The crystals are made using the following steps:

- step 1** 50.0 cm³ of 0.200 mol/dm³ dilute sulfuric acid is heated in a beaker. Powdered copper(II) oxide is added until the copper(II) oxide is in excess. Aqueous copper(II) sulfate is formed.
- step 2** The excess copper(II) oxide is separated from the aqueous copper(II) sulfate.
- step 3** The aqueous copper(II) sulfate is heated until a saturated solution is formed.
- step 4** The saturated solution is allowed to cool and crystallise.
- step 5** The crystals are removed and dried.

Calculate the maximum mass of copper(II) sulfate crystals, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, that can form using the following steps.

- Calculate the number of moles of H_2SO_4 in 50.0 cm³ of 0.200 mol/dm³ H_2SO_4 .

..... mol

- Deduce the number of moles of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ that can form.

..... mol

- The M_r of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is 250.

Calculate the maximum mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ that can form.

..... g
[3]

- (d) State **one** observation that indicates the copper(II) oxide is in excess in **step 1**.

..... [1]

- (e) **Step 1** is repeated without heating the dilute sulfuric acid.

All other conditions are kept the same.

The rate of reaction decreases.

Give a reason why the rate of reaction decreases. Explain your answer in terms of particles.

.....
.....
.....
.....
..... [3]

- (f) Name a substance, other than copper(II) oxide, that can be added to dilute sulfuric acid to produce copper(II) sulfate in **step 1**.

..... [1]

- (g) Name the process used to separate excess copper(II) oxide from aqueous copper(II) sulfate in **step 2**.

..... [1]

- (h) Suggest what is meant by the term *saturated solution* in **step 3**.

.....
.....
..... [2]

- (i) The phrase 'heating to dryness' means heating until no more water is given off.

Explain why aqueous copper(II) sulfate is **not** heated to dryness in **step 3**.

.....
..... [1]

[Total: 18]

5 This question is about electricity and chemical reactions.

- (a) The electrolysis of concentrated aqueous potassium bromide using graphite electrodes forms:
- hydrogen at the cathode
 - bromine at the anode.

The electrolyte becomes aqueous potassium hydroxide.

- (i) State what is meant by the term electrolysis.

.....
..... [2]

- (ii) State why graphite is suitable for use as an electrode.

..... [1]

- (iii) Write an ionic half-equation for the formation of hydrogen at the cathode.

..... [2]

- (iv) Name the type of particle responsible for the transfer of charge in the conducting wires.

..... [1]

- (v) Name the type of particle responsible for the transfer of charge in aqueous potassium bromide.

..... [1]

- (vi) State the names of the products formed when electricity is passed through **dilute** aqueous potassium bromide using graphite electrodes.

at the anode

at the cathode

[2]

- (b) Bauxite is an ore containing aluminium.

Aluminium is extracted by electrolysis of purified bauxite in molten cryolite using carbon electrodes.

- (i) Name the aluminium compound in purified bauxite.

..... [1]

- (ii) State **two** reasons why cryolite is used in this electrolysis.

1

2

[2]

- (iii) The anode is made from carbon.

Explain why the carbon anode has to be replaced regularly.

.....
..... [1]

- (c) Hydrogen–oxygen fuel cells can be used to produce electricity in vehicles.

- (i) Write the symbol equation for the overall reaction in a hydrogen–oxygen fuel cell.

..... [2]

- (ii) State **one** advantage of using hydrogen–oxygen fuel cells instead of petrol in vehicle engines.

..... [1]

[Total: 16]

5 This question is about electricity and chemical reactions.

(a) Aqueous copper(II) sulfate is an electrolyte.

The electrolysis of aqueous copper(II) sulfate using inert electrodes forms:

- copper at the cathode
- oxygen at the anode.

(i) State what is meant by the term electrolyte.

.....
..... [2]

(ii) State the term given to the Roman numeral, (II), in the name copper(II) sulfate.

..... [1]

(iii) State what happens to the colour of the aqueous copper(II) sulfate as this electrolysis progresses.

..... [1]

(iv) Write an ionic half-equation for the formation of copper at the cathode.

..... [2]

(v) Give the formula of the ion that forms oxygen at the anode.

..... [1]

(b) The electrolysis of aqueous copper(II) sulfate is repeated using **copper** electrodes.

State what happens to the anode.

..... [1]

(c) Spoons can be electroplated with silver.

(i) Name the substances used as:

the anode (positive electrode)

the cathode (negative electrode)

the electrolyte.

[3]

(ii) State **two** reasons why spoons are electroplated.

1

2

[2]

- (d) Hydrogen–oxygen fuel cells can be used to produce electricity to power cars.
Petrol produces carbon dioxide and carbon monoxide when it powers cars.

- (i) State **one** adverse effect of carbon dioxide and carbon monoxide.

carbon dioxide

carbon monoxide [2]

- (ii) State **one** disadvantage, other than cost, of using hydrogen–oxygen fuel cells to power cars compared to using petrol.

..... [1]

[Total: 16]

4 This question is about lead(II) chloride, PbCl_2 .

(a) A student prepares a sample of insoluble lead(II) chloride, PbCl_2 , by mixing aqueous solutions of **two** salts in a beaker.

(i) Identify **two** soluble salts suitable for making lead(II) chloride when mixed together.

.....
..... [2]

(ii) Write the ionic equation for the formation of lead(II) chloride by mixing aqueous solutions.

Include state symbols.

..... [3]

(iii) List the steps the student should take in preparing a pure sample of lead(II) chloride from the mixture in the beaker.

.....
.....
..... [3]

- (b) The student carries out an electrolysis experiment on molten lead(II) chloride using the apparatus shown in Fig. 4.1. Chlorine gas forms at the anode and escapes from the apparatus.

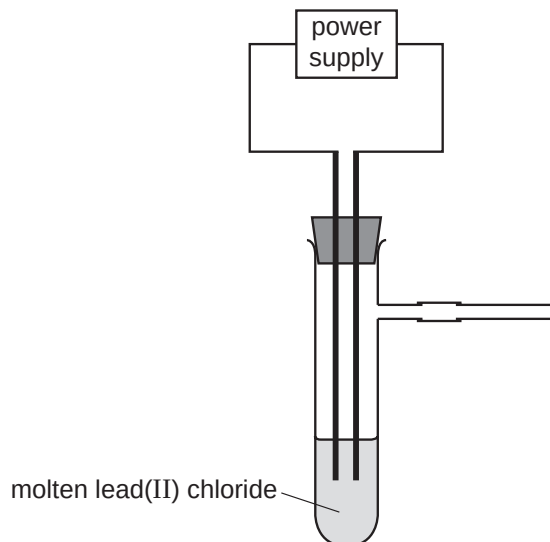


Fig. 4.1

- (i) Explain why lead(II) chloride needs to be molten before it will conduct electricity.

.....
..... [1]

- (ii) Write the ionic half-equation for the reaction occurring at the anode.

..... [2]

- (iii) State the test for chlorine gas.

test
observations [2]

- (iv) Describe what is observed at the cathode.

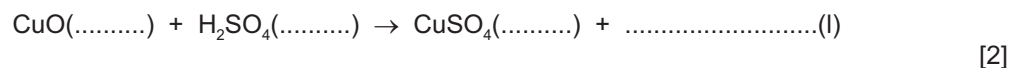
..... [1]

[Total: 14]

3 Copper(II) sulfate has the formula CuSO_4 . Aqueous copper(II) sulfate is a blue solution.

A sample of aqueous copper(II) sulfate is made by adding excess copper(II) oxide, CuO , to hot dilute sulfuric acid, H_2SO_4 .

(a) Complete the symbol equation for this reaction. Include state symbols.



(b) State **one** observation which shows that copper(II) oxide is added in excess.

..... [1]

(c) Describe how aqueous copper(II) sulfate can be separated from the reaction mixture.

..... [1]

(d) Crystals of hydrated copper(II) sulfate can be obtained from aqueous copper(II) sulfate by crystallisation.

(i) State what is meant by the term hydrated.

..... [1]

(ii) Write the formula of hydrated copper(II) sulfate.

..... [1]

(iii) Describe how this crystallisation is done.

.....
.....
..... [2]

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(e) Aqueous copper(II) sulfate undergoes electrolysis using graphite electrodes.

(i) State why aqueous copper(II) sulfate conducts electricity.

..... [1]

(ii) Give **two** reasons why the electrodes are made of graphite.

1

2

[2]

(iii) Describe how the appearance of the electrolyte changes during the electrolysis of aqueous copper(II) sulfate.

..... [1]

(iv) Describe what is seen at the cathode during the electrolysis of aqueous copper(II) sulfate.

..... [1]

(v) Write the ionic half-equation for the reaction at the anode.

..... [3]

(vi) State **two** differences seen if the electrolysis is repeated using copper electrodes instead of graphite electrodes.

1

2

[2]

[Total: 18]



2 Aluminium is manufactured by the electrolysis of aluminium oxide.

(a) State the name of the main ore of aluminium.

..... [1]

(b) Name the substance mixed with aluminium oxide to reduce the operating temperature of the process.

..... [1]

(c) Explain why the molten mixture in (b) conducts electricity.

..... [1]

(d) Table 2.1 contains some information about the processes which take place at the anode and the cathode.

Table 2.1

anode	cathode
$2\text{O}^{2-} \rightarrow \text{O}_2 + \dots\dots\text{e}^-$	

(i) Complete Table 2.1:

- Write the number of electrons needed to balance the ionic half-equation for the reaction at the anode.
- Write the ionic half-equation for the reaction at the cathode.

[3]

(ii) State why the process at the anode is an oxidation.

..... [1]

(iii) Oxygen is formed at the anode.

Explain why the main gas given off at the anode is carbon dioxide and **not** oxygen.

.....
 [2]

(e) State why aluminium is used in food containers.

..... [1]





(f) Aluminium reacts with fluorine to form the ionic compound aluminium fluoride.

Complete the dot-and-cross diagram in Fig. 2.1 of the ions in aluminium fluoride.

Give the charges on the ions.

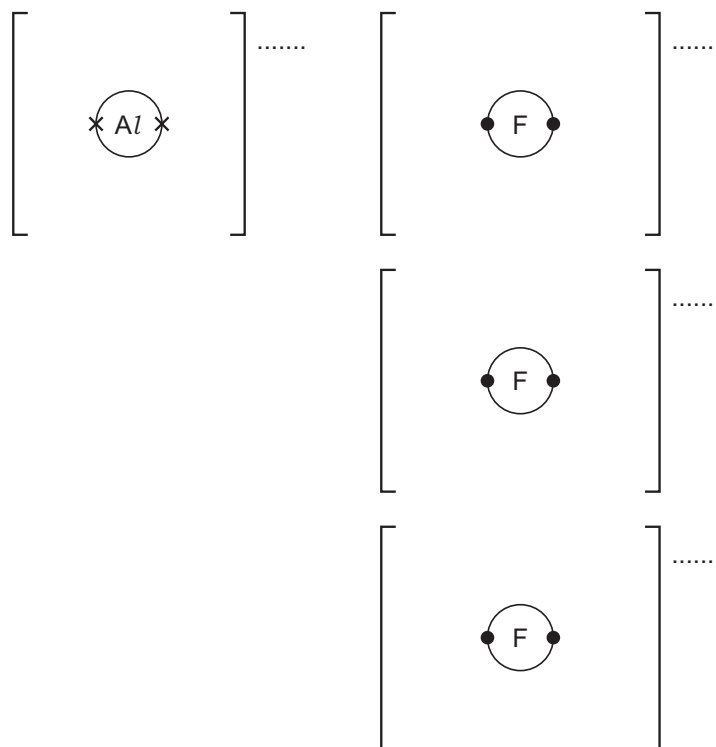


Fig. 2.1

[3]

[Total: 13]



2 This question is about electrolysis.

(a) State the meaning of the term electrolysis.

.....
..... [2]

(b) Table 2.1 gives some information about the electrolysis of two electrolytes using graphite electrodes.

Table 2.1

electrolyte	anode (positive electrode)		cathode (negative electrode)	
	observation	name of product	observation	name of product
concentrated aqueous potassium iodide			bubbles of colourless gas	
aqueous copper(II) sulfate	bubbles of colourless gas	oxygen	pink-brown solid	

(i) Complete Table 2.1. [4]

(ii) Oxygen is produced at the anode by the electrolysis of aqueous copper(II) sulfate.

Write the ionic half-equation for this reaction.

..... [2]

(c) Aqueous copper(II) sulfate is electrolysed using copper electrodes instead of graphite electrodes.

(i) Explain why the mass of the anode decreases during this electrolysis.

..... [1]

(ii) Name the product formed at the cathode.

..... [1]

(iii) State what change, if any, is observed in the appearance of the aqueous copper(II) sulfate.

..... [1]

[Total: 11]



3 This question is about electrolysis.

(a) Complete the definition of electrolysis by filling in the missing words.

The decomposition of compounds when
or by the passage of an electric current.

[3]

(b) Electrolysis of aqueous copper(II) sulfate is carried out using the apparatus shown in Fig. 3.1. The electrodes are made of platinum.

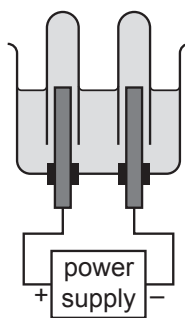


Fig. 3.1

(i) State whether the mass of the cathode increases, decreases or remains the same when platinum electrodes are used.

..... [1]

(ii) Describe the change in appearance, if any, of the electrolyte when platinum electrodes are used.

..... [1]

(iii) Describe what is seen at the anode when platinum electrodes are used.

..... [1]

(iv) Write the ionic half-equation for the reaction at the anode.

..... [3]

(c) The electrolysis is repeated using copper electrodes.

(i) State whether the mass of the cathode increases, decreases or remains the same when copper electrodes are used.

..... [1]

(ii) Describe the change in appearance, if any, of the electrolyte when copper electrodes are used.

.....[1]

(d) Aluminium is extracted from its purified ore by electrolysis.

(i) Name this ore of aluminium.

..... [1]

(ii) Name the substance added to this process to reduce the operating temperature.

..... [1]

(iii) Explain why the anodes need to be continually replaced during this process.

.....

..... [2]

[Total: 15]

3 This question is about aluminium and other metallic elements.

- (a) Aluminium is extracted from purified bauxite by electrolysis. Fig. 3.1 shows a diagram of the electrolytic cell used.

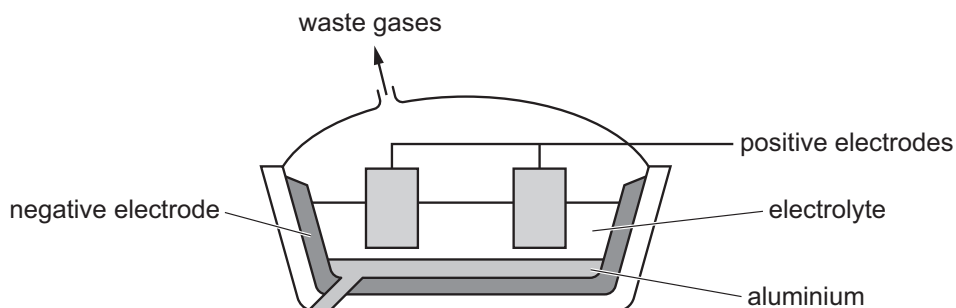


Fig. 3.1

- (i) State the name of the aluminium compound present in bauxite.
..... [1]
- (ii) Name the substance that purified bauxite is dissolved in before electrolysis is carried out.
..... [1]
- (iii) Name the substance that is used for the positive electrodes.
..... [1]
- (iv) Write the ionic half-equation for the production of aluminium at the negative electrode.
..... [2]
- (b) Aluminium is above zinc in the reactivity series.
- (i) When a piece of aluminium foil is added to aqueous zinc sulfate, a reaction is **not** immediately observed even though aluminium is above zinc in the reactivity series.
Explain why a reaction is **not** immediately observed.
.....
..... [1]
- (ii) After a few minutes, a displacement reaction occurs between aluminium and aqueous zinc ions.
Complete the ionic equation for this reaction.
$$\dots\text{Al} + \dots\text{Zn}^{2+} \rightarrow \dots\text{Al}^{3+} + \dots\text{Zn}$$
 [1]



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(iii) State the oxidation number of zinc in:

Zn²⁺

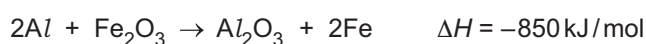
Zn.

[2]

(iv) Define the term reduction in terms of oxidation number.

..... [1]

(c) Aluminium reacts with iron(III) oxide. The equation is shown.

(i) State the meaning of the symbol ΔH .

..... [1]

(ii) State what can be deduced about the reaction from the negative sign in $\Delta H = -850 \text{ kJ/mol}$.

..... [1]

(d) Gallium has many similarities to aluminium. Gallium(III) oxide is an amphoteric oxide.

State what is meant by the term amphoteric.

.....

..... [1]

(e) The reaction of gallium(III) oxide, Ga_2O_3 , with aqueous sodium hydroxide forms a salt which contains the negative ion GaO_2^- . The only other product is water.

Write a symbol equation for this reaction.

..... [2]

(f) Deduce the formula of:

• gallium(III) bromide

• gallium(III) sulfate.

[2]

[Total: 17]





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